Umut Onur Yaşar

Applied AI Research Engineer

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Summary

Applied AI Research Engineer with a background in Electrical & Electronics Engineering. Specializes in deep learning, model compression, and computer vision, with hands-on experience spanning knowledge distillation, open-source contribution to production ML libraries, and end-to-end research from experiment design to publication.
Stack: PyTorch · CUDA · Python · Linux · Git

Publications

2026 **Student Capacity Moderates Knowledge Distillation Effectiveness: A Systematic Study Across ResNet Teacher-Student Pairs on CIFAR-10**, *arXiv:2605.31191*
Systematic ablation study comparing Logit-KD and Feature-KD across three ResNet teacher-student capacity pairs on CIFAR-10 (3 seeds, mean ± std reported). Key findings: (1) student capacity — not teacher-student accuracy gap — is the primary moderator of KD effectiveness; (2) a gradient clipping bug in Feature-KD projection layers produces systematically misleading comparisons; (3) corrected Feature-KD matches or outperforms Logit-KD in two of three pairs.
`arxiv.org/abs/2605.31191` · `github.com/umutonuryasar/kd-capacity-gap`

Projects

2026 **Knowledge Distillation on CIFAR-10 (CS229 Final Project)**
Original implementation underlying the arXiv publication above. Compared Logit-KD and Feature-KD under controlled conditions; post-submission bug analysis led to the corrected study.
PyTorch · CIFAR-10
`github.com/umutonuryasar/kd-cifar10`

2026 **Real-Time Object Detection — RT-DETR on COCO**
Investigated transformer-based real-time detection architecture. Analyzed speed-accuracy trade-offs under constrained hardware (RTX 3050, 16 GB RAM).
PyTorch · COCO · RT-DETR
`github.com/umutonuryasar/rt-detr-kd`

2023 **Real-Time Object Detection — YOLOv8**
Built a live-video inference pipeline using YOLOv8n and OpenCV, with class label and confidence score visualization.
Python · Ultralytics · OpenCV
`github.com/umutonuryasar/Real-Time-Object-Detection`

Experience

May 2024 – Feb 2025 **AI Trainer**, *Outlier*, Ankara, Turkey (Remote)
- Contributed to RLHF fine-tuning pipelines for large language models.
- Worked on reward modeling, prompt optimization, and synthetic data generation.
- Collaborated with engineering teams on model safety and usability improvements.

Aug 2015 – Sep 2015 **Computer Vision Intern**, *HAVELSAN*, Ankara, Turkey
- Developed a flight simulation interface in Python.
- Implemented computer vision modules using OpenCV.
- Conducted research on Robot Operating System (ROS).

Jun 2014 – Jul 2014 **Avionics Intern**, *TAI (Turkish Aerospace)*, Ankara, Turkey
- Built a MATLAB-based image recognition system for helicopter landing zone detection.
- Created 3D models of helicopter electrical cable layouts using CATIA.

Education

2011–2020	**B.Sc. Electrical & Electronics Engineering**, *Abant İzzet Baysal University*, Bolu, Turkey

Technical Skills

Frameworks	PyTorch, TensorFlow/Keras, OpenCV, Ultralytics, NumPy, scikit-learn, Hugging Face (PEFT, Transformers)
Optimization	Knowledge distillation, quantization, pruning, ONNX export, TensorRT
Programming	Python, MATLAB, C, C++, Bash/Shell
Tools	CUDA, Git, Docker, Linux
Domains	Deep Learning, Computer Vision, Object Detection, Model Compression, RLHF
Languages	Turkish (Native), English (Full Professional Proficiency)

Certifications

IBM	AI Engineering Professional Certificate
IBM	Applied AI Professional Certificate
Google Cloud	Integrate with Machine Learning APIs
Google Cloud	Perform Foundational Data, ML, and AI Tasks in Google Cloud
Google Cloud	Explore ML Models with Explainable AI
Coursera	Machine Learning (Stanford / Andrew Ng)